

ActInf Livestream #031.1 ~ "Non-equilibrium Thermodynamics and the Free Energy Principle in Biology"

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hello and welcome everyone to actin flab live stream number 31.1 yes we are live so today it's october 19th 2021 and we're going to be having a discussion on the paper non-equilibrium thermodynamics and the free energy principle in biology so welcome to the active inference lab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams at this short link you can see the past and upcoming live streams here we are at the end of october 2021 in paper number and today we're going to be having an awesome group discussion and then next week we hope to have one or both of the authors join us so we'll be taking a lot of notes today and bringing up good things to talk with them on and all of these are participatory live streams so if you want to join or if you want to co-organize with us just let us know today in 31.1 we're just going to be learning and discussing this paper of palacios and colombo and we're just putting out some perspectives giving some affordances for others to join in hopefully learning along the way so we have a few questions that blue and others have prepared maxwell also prepared some notes to talk about and we're just pretty much going to look through the questions that we have already written up we'll add more anyone in the youtube live chat who wants to ask a question they're more than welcome to do so and we'll just hop around wherever we need to go we'll start off with the introductions so we'll just say hi and maybe something that we liked about the paper or something that stuck with us or something that brought us to this conversation so i'm daniel i'm a researcher in california and i think something i'm excited about today maybe would be to look at this intersection of biology and thermodynamics talk a little bit about the different ways we can approach that intersection this paper has some formalisms but not all the formalisms there

weren't figures but there's so many graphical intuitions related to free energy landscapes and some other pieces so it's just kind of cool to think about what are the different ways that we come to learn about the intersections of these cool ideas and i'll pass to stephen hello yes stephen i'm in toronto in canada um pleased to be here again and i'm interested in this paper the general topic of ergodicity comes up and um i'm interested to see i often see these papers which have sort of free energy principle focus rather than say active inference focus so um just to see how in that when that tree gets shaken how that helps or hinders the sort of more processed theory work which is probably closer to where i'm uh interested in but this is all very interesting thanks i'll pass it over to maxwell hey i'm maxwell ramstead i'm the ceo of nested minds network an active inference ai startup and i am a fellow at the welcome center for human neuroimaging ucl in london i'm looking forward to discussing this paper i think there's a lot to disagree with and i'll try to be collegial in my criticism beautiful what more could we ask for let's start with uh the big question and of course either of you if you have a different take or there's other big questions but this big question which is in this paper as well as the broader literature what fundamental ideas from physics are utilized in free energy principle and active inference and how and just what are the implications or trade-offs associated with the use of this idea does it lend credibility to a theory to be grounded in physics what does it mean to a theory for it to be grounded in a certain physical idea does it constrain does it prevent us from making the kinds of conclusions that we might want to draw about biological systems so that's kind of the big question do either of you have any other sort of big questions that this raises or what is the alignment of like the vector of this paper with bigger ongoing threads in our research community well i mean i guess one outstanding question for me is uh what is the horizon of relevant literature in terms of time course that one needs to consider before you know trying to engage with and criticize this material it's something that comes up i think and it might be useful to discuss maybe before we go into the specifics there just when we're engaging in interdisciplinary or transdisciplinary scholarship or we're diving into fields where we might not have taken courses to the undergraduate or higher level just it's a really important question how do we make appropriate claims you know how do you take a little bit of a of a dip into the rabbit hole and come out with something that's useful rather than just being more confused than when you made the initial search so these are big questions any thoughts are just how do we determine what literature are relevant like if we wanted to write a paper on something related to active inference or just in general what do we do do we do keywords do we only look at the last five years are you asking the question

rhetorically or do you want us to chime in oh maybe steven first with a raised hand but then just either a mute steven and then yes steven unmute and then continue i'm probably not the best person because i keep digging and digging and digging but um i think it's a good question i i i'm from a chemistry background a long way back and i always find it interesting as well because you know a lot of stuff that's used is or developed it comes from chemistry actually um uh and uses physics but uh you know the the the like you say that we're in this kind of need to approximate and an approximation what like there's so many things that come up that are like new new ways of describing stuff which makes total sense like i i've read through the free energy principle and kristin's work about the talk about approximation science and that science is ultimately an approximation and then you know it takes me a couple of days to then get my head around that and it's like yeah that totally makes sense but the implications of all of that are huge the implications of being in non-equilibrium states itself disrupts a lot of the accuracy of of certain ways that you can roll out things from physics and are you rolling out into chemistry which is normally equilibrium chemistry are we rolling it out straight into biology are we rolling it out into something slightly different with all this modelling so i think it does cause some challenges in terms of discussions because it's easy for other people to start telling the free energy principle and active influence community what they are and then start telling them and framing what they're not and by which point it's kind of hard to have a discussion because it's like the straw man's been set up and i think that can be a little bit challenging for some conversations to move forward because now you've got a complex idea trying to first of all counter the fact that other people's paradigms have been challenged uh so i think that itself um i don't know if that's a literature thing or whether that's a cognitive thing but i think that that is part of the challenge that seems to be faced agree steven thanks for that and it's maybe a good segue maxwell we can just start to hear some of your thoughts and all be typing some notes and maybe asking some follow-ups and unpacking because i know you'll have a lot to say on this so maybe just go for it and i'll just type along and we'll see where we go i mean um i guess that probably the thing that i find most puzzling about the paper is that it claims to be the first in the literature essentially to be looking at the connection between the physics of the free energy both physics and the free energy principle i find this puzzling because it seems to me that the connection between physics and the free energy principle has been really maybe thus the focus of active research in our community for like four years or so um so i mean you know the carl published or at least pre-printed his uh particular physics monograph in january um and you know prior to that we had all been working on i mean

effectively the connections to physics so uh i find this problematic because the work that we're going to discuss today focuses on you know older formulations of the free energy principle where the connections to physics were maybe less obvious i mean as most of us in the community know now the formal foundations of the free energy principle have been subject to extensive debate over the last you know three years or so kind of martin beale and colleagues kind of started um started us off in this direction and you know uh since um martin and uh you know a few other uh groups of uh produced uh you know have stimulated interesting discussion through their very interesting and technically rigorous critiques of the free energy principle but really it seems to me that this has been like a focus of our discussions for the last several years the connections to physics and so on um you know and where this has gotten us is uh the bayesian mechanics for stationary processes paper that was recently released um which accounts for the critical remarks that have been raised and you know it so really where we're at now is that we have uh formulations that start from non-equilibrium statistical mechanics and information geometry um and end up you know saying well all systems that have stationary density dynamics and a markov blanket will look as if their internal states are uh you know modeling in a loose sense or in a more technical sense encoding the parameters of probability distributions or bayesian beliefs about external states that's sort of where we're left with and you get you get to this from the physics so i just found that the um the remark was a bit puzzling and the choice of literature was also i think a little puzzling for the the same reasons i mean i i have more specific thoughts but i mean you know this is sort of for your first bullet point in response to the last one that we added what's the relevant time horizon i feel like a rapidly developing theory like the active inference framework and the free energy principle i mean if you're going to be producing technical critical material you should probably be working from the most recent versions i mean i find it puzzling for instance that the authors spend so much time talking about ergodicity when the word ergotic doesn't appear or just just about never appears in the particular physics monograph so anyway i just find that there's a kind of disconnect between what's presented here and the actual work on the physics of that we have seen and uh you know i think that leads to several of the arguments uh being proposed uh being a bit um missing their mark to put it diplomatically thank you for the concise summary maxwell so steven with a raised hand and then we'll continue could i ask a little question as well that relates to this is in the very first part of the abstract it talks about the um inevitable and emergent property of any ergodic random dynamical system at non-equilibrium steady state that possesses a markov blanket so that's a quote from the paper from

2013. the thing is it says before that according to the free energy principle life is an in and then it quotes inevitable and emerging property which is the quote but i'm not sure that life is ever talked of as an inevitable and emerging property of any random dynamical system in terms of how it's referenced and i'm wondering if there's a bit of a confusion in the paper there where it talks about this that quote but makes it as if that's what life that life is an emergent and an emergent property and i think it's um i think what we're talking about is adaptive reduction of surprise um or bayesian um inference properties is an inevitable and emerging property but i don't know if that would be life but i don't know what your thoughts are on that or whether i'm misreading that that reminds me of one of my favorite papers what is life answering schrodinger's question by maxwell at all in 2018 um because maybe you could even say something to this what is the relationship between reducing surprise and life and do these recent technical developments in fep help bridge the gap between inanimate and living cognitive or computational processes and where does physics play into that that's that's very interesting i mean i think the the fep itself is not a theory of life just in and of itself right what is what it is it's a theory to understand systems that are able to maintain themselves in non-equilibrium regimes effectively uh so or i mean more precisely you know stationary systems with a markov blanket which is to say the same according to the theory um so i've gone on the record as saying that the free energy principle can allow us to construct a theory of life but the claim isn't just that the free energy principle in and of itself allows you to do that i think you know the fep needs to appeal to auxiliary hypotheses from you know biological and other sciences to really do that i mean it's a theory of thing-ness physical systems that interest us and a subclass of those systems are biological systems um so i mean it is an aid in the formulation and probably the foundation of a formulation of uh living systems so so long as we agree that living systems are a subclass of persisting systems and i mean there's some talk in this paper that maybe this persists living systems of persisters might not be the right way to think about life but yeah i think it acts as a foundation i mean but then you need to add a lot more to turn that into a theory of life i mean if you read my schrodinger paper carefully for example what we propose is a variational approach to neuroethology uh we're not saying that the free energy principle in and of itself can just offer this like the free energy principle has to use as it has to be used as a foundation from which you can build these uh variational approaches to neuroethology to niche construction to ecology or what have you so i mean but you know that that might just be uh picking it nitpicking so like i uh you know well i don't want to double down on that too much it's good points and it's almost related to the definition or the

distinction between like a discipline and a trans discipline if fep is a discipline versus if it's connecting other disciplines variational neuroethology it's right there in the title we're connecting across different areas and the framework of thinkness is one of those enabling features that helps it be an integrative theory but potentially it doesn't fulfill the disciplinary uses of theory so that's one interesting idea um i guess there's many other ways to go or things to ask i guess i'm just i'm curious again like maxwell what do you think in the last few years how have we clarified or learned more about where active inference and fep are related to each other so how was that thought of in five years ago maybe when you were in grad school and then what do you think is sort of contemporary greetings blue in terms of the distinction between active inference and fep and then welcome blue and you can add anything you'd like and i know some people want to mark the difference between the two i think personally i think it's more of a sociological thing uh because no one or of just about no one is contesting the validity of active inference models of cognitive systems but there's a lot of pressure from different circles on the free energy formulation so i mean it might be a sociological thing where it's just easier to say oh i don't work on this more you know metaphysically sounding thing i just work on computational neuroscience and that's straightforward uh i think there's no warrant for that from a mathematical point of view you should just for example look at the way that things are derived in the bayesian mechanics paper yeah i i feel like uh active inference just falls out of the the core bayesian mechanics i mean i'd recommend everyone read through that bayesian mechanics paper it's really fantastic it's very clear one of the big differences between it and its previous free energy papers is that uh it's authored by or at least there are two authors out of the four on the paper who are proper mathematicians working on you know the mathematics of non-equilibrium statistical mechanics effectively so uh you know that paper really uh i think meets the standards of rigor in mathematics you know you have an existence proof for the synchronization manifold that's sort of at the core of the whole formal construct the thing that links internal and external states and then you have a construction of it i mean it's very straightforward and having having established you know the kind of core formal apparatus it's unfolding and by the end you know you what you do is ask kind of well all right let's split the blanket states up into sensory and active states and active inference just falls out of you know making the active states subject to the dynamics that those are unfolded throughout the paper so in my view the connection is immediate yeah uh i i i will grant that you know they're some of the computational models that we're using are a bit more um they look a lot more like just vanilla

you know computational neuroscience models uh you know their palm dps and stuff like that but but what the math that underwrites it all is the math of the free energy principle so i don't really i don't really think the two can be separated thanks maxwell and um the basic mechanics paper we discussed in live stream number 26 so it's um it is a lot to keep up with sometimes but that was a very fascinating discussion so blue and then anyone else i was wondering if that was bayesian mechanics for stationary processes the one we did yeah that's precisely it yeah um so i sorry i'm late on the scene but um i am curious about whether is the fep a theory of life or does it allow us to construct one and does it only apply to life or can we take it in this kind of pan psychic direction where even molecules like want to minimize their free energy and self-assemble to become structures that's a very good question um i would say that the free energy principle is not a metaphysics so it's it's a physics so it's a formal framework may or may not give us grip on the systems that we're interested in so i mean you know i and i i know there is some philosophical literature that presents it is more like you know a metaphysical statement about what life is but i think it's more useful to think about the bayesian mechanics as just an extension of mechanics effectively you know the the same way that you know statistical mechanics and quantum mechanics are flavors of mechanics the bayesian mechanics is also you know and i realize that it you know it's a little deflationary uh it's a more instrumentalist reading of the free energy principle but it's not completely deflationary in the sense that it the model says that the best model of livings and other existing systems is as a model of their environment there's a kind of double instrumentalism at play i think in in the free energy stuff to answer your question a bit more directly i think it is it can be the fep can be used to construct the theory of life in and of itself it's a theory of thing-ness but yeah it's a great set of questions to discuss and i mean this is only my view like i don't mean to settle this once and for all there are people who want to defend a more um metaphysical view on this um i guess the question is how far do you want to take the commitments uh of the free energy principle i mean you know the there are kind of two irreducible assumptions that go into the math there's stationarity which by the way is not the same as ergodicity so you know uh i think a whole section of this argument just doesn't apply at all anymore the one presented by colombo and palacios and markov blankets so i mean you know if if you think that you know systems that exist at some scale have to be stationary and you think that lit you know systems that exist really do have a markov blanket then you can be realist about the assumptions but i kind of feel that it works whether or not you take the additional philosophical step of committing hardcore to the realist assumptions you know it works just it works as an you

know an as if theory if you will i mean the whole thing tells you like living sorry uh systems that uh existed on the at non-equilibrium study stated have and have a mark all blanket will look as if they're inferring external states and i mean um yeah it i think this this looks as if stuff is really important and i realized that you know for for some of us it may be a little bit deflationary uh but i don't know if more is warranted from the math i think that you know there's a yeah we just need to be kind of careful thanks for identifying those two key pieces and maybe we'll have more to think about with the stationarity and ergodicity so blue with the raised hand and then stephen so just a couple things like when i say pan psychicism i wasn't trying to go into like the metaphysical realm but because it's physics because it's not metaphysics can it then apply to life and non-life like can we just go all the way down yeah and so in that kind of way i wasn't really implying as a metaphysical construct but more in like the direct does it broadly apply to non-life as well as life yeah and and then also um just a couple more things systems have a markov blanket like maxwell that's you being entirely realist like i don't have a markov blanket but can i be modeled with a markov blanket probably um so so i don't i don't know like you know i don't curl up in my markov blanket at night when i go to bed comfy and then just lastly like going back to the paper um this like so i don't know that was one of the main questions that i have for the authors is this concept of they say like systems are in dynamic equilibrium i don't think that the non-equilibrium steady state density is the same as dynamic equilibrium at all and that was one of the main questions that i raised and like dynamic equilibrium really is not it's a chemical process like of a reversible chemical reaction so i i don't i i really kind of missed that that was one of the key things in the paper that i just didn't understand at all yeah i uh i mean i i very much agree with you on the points that you're saying the requests that you're raising here um so i mean regarding the pan psychicism thing i mean yeah you can from physics now kind of say that any system that exists kind of looks as if it has a mind uh you know then you might not want to go full on the metaphysics of pan psychicism but it certainly warrants that kind of claim i mean in my view um this paper is making uh a few critical mistakes uh and this is echoed a lot in um you know a lot of the literature coming from more like four e people writing about this i'm just gonna cite you know two sentences from a paper pre-printed this year uh i won't i won't name names but you'll see the style of what i mean to be as fair as possible we'll do this as in criticize the fep by attending to the more recent published versions and variants of these claims and then the next section starts literally the next sentence the work presented in first in is central to practically all the proposals comparing the fep and in action and it just goes on and stuff and

it's i i sort of feel like yeah the physics foundation has moved forward like you know blue you're pointing it out it's all about non-equilibrium steady non-equilibrium steady-state dynamics and you know stationarity that's importantly different from ergodicity uh you know the one of the nice things about the the bayesian mechanics paper is that it it abstracts away from those assumptions i mean to be clear the free energy principle at its core never really required that the systems that your study be ergodic we assume an ergodic density in some of the earlier publications because it allows us to finesse one step in the derivation substituting the time average for the ensemble average and one of the really core derivations but you know as uh carl himself had been pointing out for a long time you could finesse the math into you know something else like there's no reason why you have to stick with the ergodic assumption it was used to finesse one step of the derivation and you know what lance de costa and a company have done in this bayesian mechanics paper with carlos precisely to do the laborious work of actually generalizing the math to non-ergodic systems so i mean i think that's really important the bayesian mechanics paper um and you know we're in discussions with the people active discussions of the people who've produced the critical material and i mean at this point i think everyone essentially agrees that lance's new derivations respond to you know the the critiques by aguirre colleagues that came out earlier this year and also the beel at all critique remember the the core of the beel at all critique is that uh for ou processes in general of the sort that carl was using you can't assume that the synchronization manifold will exist between internal and external states in fact what martin and company have showed is the for a generic ou process you can't just you it does not exist um but you know that that doesn't mean that you can't find very broad conditions under which the synchronization manifold exists and that's what lance and company have done in this new aguirre's critique aguero at all this is the sussex group also did a very interesting analysis of the math as it figured in the uh particular physics paper you know showing that actually the the precise assumptions that carl uses in the paper would restrict the scope of the fep to a rather narrow class of systems but again lance just rederived all of the equations so the formalism is now effectively different and what you can really see from the math is that it applies to non-linear systems and to non-ergodic systems and this leads me to another point um i think the the the assumption of non-equilibrium steady state is often not understood in the way that it should so non-equilibrium steady state is a characterization of the density dynamics it is not a characterization of the system itself if i if i say that a system is at non-equilibrium steady state i'm not saying that it is steadily in some state you know that it just that it occupied what i'm saying is that the the probability landscape

that underwrites its dynamics is stationary or has stopped evolving um those are two very very different claims so it doesn't follow from the fact that a system has a nest density that it is history-less it doesn't follow from you know the the existence of the next density that uh there should only be one attractor like you can have metastability in an fep system you can have non-linearity in an fep system you can have non-ergodicity in an fep system yeah i mean i i think that means you know that one of the three sections the critical sections in this paper are just completely off the mark and you know there's this new paper by some of the tenors of the inactive approach on precisely this kind of stuff as well unfortunately the same mistakes are made um you know ergodicity in in the context of the fep i mean really just means that the the quantities that you're uh observing or measurable um it means that you know all the average of your measures converges on average to a measure of the average you know or of the mean that you're actually trying to calculate in a heuristic sense so it never meant that systems don't have a history it never meant that systems necessarily converge on one attracting set and so i i really feel that that kind of takes the teeth out of most of the you know formal criticism in this paper uh the other part of the the other critical parts of this paper talk about you know the challenges of modeling biological systems using state spaces but that those sections to me don't amount to much more than just saying it would be very difficult to do this therefore it's not doable or something like that and you know i i i have a lot of friends who are computational biologists and i speak to a lot of people doing this and it's sort of like well you know some work would have to be done to write a paper to respond to this but you know what's the expression one person's uh modus tollens is another person's modest modus ponens or like you know i sort of feel like all right well you know you say this is impossible uh i know that formally you know alex alec chan's is a really cool guy and uh he's brilliant he and the sussex team uh have recently recently i mean this is two years old so it's not that recently and i feel like this should be accounted for in this paper but have recently automated uh state-space construction um in the free energy framework like we have active inference models now that can kind of automatically assemble and can handle up to it's either 10 000 or a million so you know that's a lot of modeling power and i i find the example that they discuss in a footnote there uh you know however this is on page 11 however the size of the genotype space for a protein consisting of n residues chosen from an alphabet of amino acids the size m is m to the n and then they say oh well you know it's like 20 to the 100 or 10 to the 130 possible uh states if you wanted to model like you know amino acids what they're not considering is the nested formulation and i i kind of feel like nested nest densities should have been at the core of

this argument and figure nowhere uh yeah i mean of course if you think that you can just plot all of the states of the system across all the scales at which it exists in one you know homogeneous state space you're going to run into a problem but you're not if you carefully construct these state spaces in a kind of nested fashion um you know like a friend of mine van the tower uh works on i pulled this paper up he has a really cool pre-print a deep learning approach to capture the essence of candida albicans morphologies and i mean basically there what they do is start with a population of candida which is a fungus for those of you who don't know and then you randomly generate a crispr and you introduce it into the population and you see how the population develops and you plot each populational trajectory in a 30 000 parameter hyperparameter space effectively and you see the way that like the candida self-organize into pockets depending on their you know the kind of crispers that you put into them uh i mean this is not like fringe work this is just routine computational biology where we're already you know plotting phenotypic expression in 30 000 dimensional spaces this isn't like forthcoming work to be done i mean it's really just standard work it's interesting how you know i i think they they modified the paper we had discussed this in a live stream before and they they nicely added some uh responses to some of the criticisms that i i raised in discussion i'd have to pull it out but you know i i said well you know that we're using this stuff in studies of uh cancer dynamics and you know like so uh i i wrote in a paper you know we can think of a biological system's extended phenotype as the set of extracting attracting states of a coupled dynamical system and this is supported by studies of cancer genesis and propagation uh i can link the papers and you know they say well you know these studies provide evidence that cancer can be fruitfully modeled and understood as an attractor in a dynamical system but they don't provide us with evidence that all biological systems occupy an attractor and i'm sort of thinking like well no but it you know this is a counter example to you're saying that it's not doable in a real biological system you know uh so i you know if i if i'm being snarky i feel like this is armchair biology of the worst kind kind of you're telling me that what i'm doing is impossible while i'm doing it and you're doing it by appealing to like you know an old conception of the physics uh by talking about assumptions that aren't even part of the formalism anymore i mean i i just uh i find that there's a lot to object to in this paper both in terms of the substance and in terms of the general kind of strategy of kind of gotcha in now does the free energy principle need to be discussed critically absolutely and you know i i should have caveated all this by saying that some of the most interesting and stimulating uh intellectual exchanges that i've had in my life have been with you know the people creating

critical material uh but you know the the people that are generating this interesting critical material are actually engaging with the theory at its formal core you know i i sort of feel like some of these things would have just been cleared up with you know a few conversations and looking at the more recent material but now i feel like i'm just kvetching at this point thank you maxwell that's what these discussions are for and hopefully bringing into the open it's just a multi-perspective scenario to give you like a specific idea why this is problematic like if if you focus on ergodicity and not on the nested nest densities then you're going to say things like well first you have to show that there's an attractor and then you have to show that the relaxation time is biologically plausible and they claim that neither of these things are done in the free energy literature when literally you know the core 40 50 pages of carl's physics monograph deal with and you know dan uh i i i know you we've we've read it all together during the last year i i don't know how you how they could have missed this i mean it's it's the physics monograph this is supposed to be their topic the core of the physics monograph is all about you know uh information length uh effectively right and relaxation times back to non-equilibrium steady-state and how all these things kind of work in a nested way to get you biological dynamics so it seems to me like they are just completely missing their target but both by you know working out on on formulations that are out of date and that introduce assumptions that are no longer at play and and also by just really ignoring the physics stuff the free energy physics stuff that they claim to be criticizing uh we need to be um just like if we were going to catch a baseball we need to be performing anticipatory knowledge work not merely reactive especially to previous uh formats and then or if it's reactive i mean you know react to the recent stuff i mean you know there's been a concerted effort to ground this in the physics of non-equilibrium systems like non-equilibrium statistical mechanics information geometry i mean if you're going to be writing a criticism of that stuff then you should probably be writing about that stuff is sort of my point and you know and by the way this often goes really badly when i point out that hey you know you're working from a you know an outdated thing they say oh you know the free energy version 25 from this morning and blah blah but i kind of feel like you know my friend if you want to write a criticism of something like you should be as charitable as possible with the target of your criticism on the one hand and you should definitely not be working on versions of this that no one in the core community is still working on and there is a dimension of like if you can't keep up then i don't know you know i don't know who benefits from this it's sort of like my my questioning it's like okay so steven and then blue yeah i think thanks for the giving us a good good insight there i i think the the

site the sociological psychological piece is present in all of this as you say and i think i've mentioned this before to you maxwell actually is the the danger of systems talking about systems i think even if you were talking to people in the international society of system sciences they often talk about systems system dynamics and everything else things complex systems kind of gets lumped into complexity right so there's a danger because i don't think carl necessarily would say fep system right and that term system i think gets you into a lot of challenges because the folk psychology of what a system is is what we see it as from the outside and you know these non-equilibrium steady states i'm wondering whether we say mark he says blankets all the way up and down but i'm i'm not sure if it's more like thing this all the way up and down is safer than systems all the way up and down because things can have all sorts of you know and they can have states and they can be what they are and the question really becomes then within all that ergodicity which unlike physics which tends to be about getting things to a certain number of decimal points of accuracy jen it's a messy thing you know i was talking about that thing about approximation science i mean again that gets your head around what that that makes suddenly you got well the main thing is you just need enough of whatever this stationary extractable data is to be able to do something with it and i'm kind of thinking that maybe because it in the same way that in chemistry if you have a beaker and you stir it up and basically the thing will have its dynamics it will move based on the energy between the different things and basically entropy is just basically the the force that's pushing it in the way that you don't expect it to it's basically you know we know it does this from this equilibrium state we know that and entropy is just the thing in the middle now no one has a bloody clue what's why that's just a black box right now but that's because they're going from now when you go to these non-equilibrium far from equilibrium chemistry state and they're held which you don't do when you mix up a beaker and just go from a to b now we don't know how many non-equilibrium states these reactions go through to be honest in those cases now but you wouldn't call the beaker a system you call it a solution you'd call it a you know you might call the vessel and the chemical plant in which it reacts a system but you just sort it kind of a beaker of stuff or things that's doing stuff and things right and you sort of see what it does and then it's like what can we do that's useful and normally what's useful is what energy shifts but in this case you've also saying the entropy can be useful but not in the way most people would the what we could say the folk systems thinkers would kind of tend to think about it so i'm just wondering if there's a danger using the word system yeah i i would agree i mean uh i think we use the term system just because uh you know all of this

grows out of dynamic systems theory in some sense carl talks with particles right like uh a a system with a markov blanket and does the free energy stuff is a particle in his account um so that yeah there is a uh so there's a there's a linguistic danger here because i agree with you this is not what people think of when they hear the word system a lot of our exchanges as you pointed out stephen have turned around that specific point i think the the term is important because especially in the context of relating the fep to like the four e approaches i think it's not appreciated sufficiently that the fep math is what happens if you take dynamical systems theory seriously um you know so when i was doing my undergrad i was reading the four e guys the fourier people because it's not just guys uh yeah the the fourier people were telling me all right well you know computer theory you only get so far in the study of cognition you have to do dynamical systems learn dst math and learn the physics of complex systems so i did and several of us did but what happens is then if you continue asking questions along that direction things arise like well how do you measure distances between states and if you start asking that kind of question inevitably information and entropy creep back into the picture because that's how you measure distances between states effectively is in gnats and other information theoretic units so i mean this is to me the great irony of this whole you know fpp versus an activism confrontation which is slowly turning into like one of the big discussions in contemporary philosophy is i don't think that the fourie people fully appreciate that the fep stuff is just a prolongation of their thinking style that if you if you really want to take seriously the core tenets of embodied cognition then inevitably you are led back to information there's no escape it's really if you want to take their own framework seriously there's no other place to go but here i unfortunately have to uh leave now thank you maxwell thank you for listening to me rant about this i'll try to be there next week um and yeah i mean uh keep kicking ass i mean dan blue uh i i just love active inference lab and i'm so grateful for the awesome work that you do keeping this community alive stephen it's always a pleasure to talk with you and i'm glad to see that you know there's uh so much uh so much like yeah you're just there all the time and i just love your uh you know your your contributions to the conversation it's always a pleasure to talk to you so a big love to everyone uh and i will be uh in touch shortly uh apologies for the snarkiness it's just this is the kind of paper that really grinds my gear i thought it was it was uh good to just point out a few of what i i think are the issues thank you maxwell see you soon goodbye this uh blue will go to you but this is uh you know it's a key difference between being an organizer and a facilitator or a participant we're here to help scaffold individuals to participate and their perspectives and uh that's the second order cybernetics so things will get

more contentious than we can even imagine and someone just has to hold down the broadcasting make sure that hands are being raised and that protocol is being respected because there are serious first level questions to resolve but we can have precision in our second level policy and that's gonna return to a question in the live chat but first blue with a raised hand thank you so i just wanted to um you know say that a lot of what both maxwell was saying and um what stephen said just really underscores the need for consistent definitions like how do we define a system and also like a consistent synthesis of the fep like historically where did it start where has it been where is it going and like what where are the landmarks like for example this basin bayesian mechanics for stationary processes paper you know has revamped the math and physics and when we're making a new paper or a new argument are we working with the current version of the model absolutely nice uh foreshadowing of some work that we're involved in but if anyone is interested in sort of scholarship and a history from an empirical perspective on fep and active they should get in touch with blue or i because this is a project that we're super keen on but it's a very important point and that's why in our dot zero we had like this slide with um the active inference lab the working ontology so we're going in a participatory way and we're working through these terms and we're finding where it's quoted in the literature how it's used in the and then we're adding notes and relevant resources and we'll be developing curriculum that helps make simple declarative uses of these terms so that we don't get lost and we'll have the terms and we'll have the specifics of literature and it'll be an awesome environment for learning applying and communicating active inference so before um stephen you want to go with a question and then we'll go to the live chat well i just add to that point you just made is i think it is really going to be important because we're seeing the confusions that can happen but also it's it's likely that there's going to be a kind of a multi ontology there might be something in this whole field that um speaks to how this changes just in the way that carl fristen talks about when something is regimes to an infomax focus versus a hamilton lease action focus versus uh you know a reward focus so there may well be this commonality and there may also be a way to process when things particles become you know derivations and you know that's going to be interesting because there may be some nexus whether it's called metaphysics or whether it's it's something more prof you know less less out there maybe it can be grounded and i think that that's starting to come up as we we hit you know that's transdisciplinarity for you right it's gonna you're gonna want to know what's in the discipline piece so maybe there will be stuff which is kind of grounded in the biology world and grounded in the physics world and is

going to talk as best it can their language right and then there's going to be stuff which is in the applied world which ultimately starts to go transdisciplinary at some point as soon as you get out and get to any sort of scale so i think that's going to be very very interesting awesome and both steven and blue and many others join us weekly in edu where we talk about this so let me read a few of the awesome comments from the live chat and ask a question from the live chat and anyone else can continue to ask a question so there's a lot of discussion on the question and yvonne said maxwell subscribe to our systems thinking course we will discuss what a system is from the standpoint of systems engineering so that's really important thanks yvonne for sharing that and indeed two of the fundamental pillars or ontologies or trans disciplines of active inference lab are active inference and systems engineering so those are the perspectives that we draw on and seek synthesis and application of and we're all learning a lot and it will be updating our generative model in a way that changes our policy selection to be able to use words like system without going down the philosophy rabbit hole every time and just having a way that we can talk about systems of interest like in dot coms our system of interest is the live stream so how do we take a role-based approach that embodies the best of our values and uses the term system and we also get all these other pieces too so good um thanks for sharing that yvonne dave shared a lot of interesting information about life and about some of the history of a few different ideas related to conceptualizations of life but the question that i wanted to get to was from michael nyc who wrote how would you model this conversation using this model so that's a question we can uh think about and i it speaks to almost like applying because online teams are certainly an application that's important for different kinds of and i'll give a first thought on this so one thing is when you said how do we model it the question is beautifully phrased because it doesn't say well what does the theory say that this conversation is that's setting up for an absolute hegemonic realist answer how would you model it is going to be inherently different perspectives because different individuals and different years are going to have access to different tools and ideas and people so they're going to model this conversation differently so that's the first piece and that's moving from that sort of absolutist modest realism into more of a multi-perspectival system what are some ways that one could model this there could be a cyber physical depiction where one is very interested as we were in our 2020 voyatkin at all in the information flows from a computational perspective like sense is the incoming states and actions are the outgoing states so in the context of a video conversation there's an audio visual feed in and there's an audio visual going out so we could use blankets and interfaces and sense and action states to talk about just the cyber

physical dimensions of the conversation another approach would be the hermeneutic or computational conversationalist approach which dave has connected back to a long lineage of work related to 1970s conversation theory and he wrote it provides a micro theory for nested mutual models focusing on the evolution and incorporation or rejection of predicates and their syntactic occurrence active inference work provides a corresponding needed macro theory focusing on secular evolution of the interchanges so one could have a communication oriented model that focuses on the semantics of the conversation one could make also as discussed that kind of a cyber physical very computational model and there's other types of models that can be made and then there's a million different implementations so that's just the first pass but it's a really important question and it helps us connect the abstractions and the framework to the minute particulars which are the system specific realizations that ground us in transdisciplinarity and application so stephen and then any other questions from the live chat or blue yeah i think they're quite useful questions those because they're that question about how can i um i mean it ties into a couple of things my first interest would be say okay why would i so there's some but the other thing is how can i i how can i implicitly also link to is available to me what information is it possible to interact with you know that idea of so we can bring in we this modeling process sort of ties us in this sort of mid ground between how we actually want to talk about all of this which i think is where the um systems engineering is really useful because ultimately you kind of have to bring it into a system of externalize knowledge which can be then communicated and moved around as in the papers and other things that can be done but it's also how can you start to understand say the process is going on in the moment so it could be measuring what we're doing in this real time process and someone tracking that it could be a case of video in it and watching back the video with an interpretation thereof it could also be you know if we were wearing sensors you know they could be just trying to model the dynamics of us as things you know there's a just trying to say how could i know what i can even trying to feed into a system so i mean that one of the classic cases that's recently there with ryan smith their work on interception they had to devise a method for recording first they did it with the heart and they changed the way the heart was measured but they also did some recordings of the stomach and the stomach region they had to find a way to get data to find out what could then be modeled was that a system they were modeling or was it almost the dynamics they were modeling and of the things you know and i think that's i think and just seeing how all the dynamics change over time and i think this is this is where maybe there's a lot of challenges um that's going to be faced because we've had

this conversation in this modeling world is jumping things into a system engineering world which isn't only a non-equilibrium philosophical world it's quite useful for applied work because it does that ability to make something a system in a way that's intentionally that and it has system of interest just like daniel was saying and then you've got but within that our system of interest is about going beyond the systems of obvious presentation you know there's stuff going on under the hood which is not necessarily um intuitively available at all so i don't know what your thoughts on that daniel but i think it's interesting yeah the only short thing i'd say is like even if the system of interest for example is the mind of someone being educated for example in the curriculum of active inference lab or the audience who we're speaking to is our system of interest or even live stream having your system of interest be the learner's mind or brain doesn't mean you're looking for the neuron level description having the system of interest to be the live stream or the audience doesn't mean that we need to know every single person or how video encoders work so it's an application oriented approach perhaps even an action-oriented approach that cuts through some of these questions or sidesteps or saves them for savoring you know after the project has been completed after we've made the changes on the real world it savors some of these truly excellent philosophical and important questions while also allowing us to act reasonably in the moment and so it's always that north star of a unified model of perception cognition and action that's so motivating so even as some of the formalisms do evolve we hope that we can respect the the spirit and the inertia of the research that we've carried out and also know when we can modularly swap in new elements of the framework so blue and then of course any questions in the live chat as a modeler how would i model this conversation really like let's talk about how to really model it um and not from a philosophical standpoint like if this is my model of the fep i have a model of the fep that exists in my brain as does daniel steven maxwell we all have a model of what the fep looks like every time we read a paper have a conversation take input from our it's the action perception loop right my model exists it's in my internal states through my senses i perceive read here something that either confirms my model provides evidence for my existing model reduces my uncertainty about my existing model so i have some evidence input through my senses then i update my model or if it doesn't jive or if it makes me realize something i mean always we're gaining information so there's this epistemic value add through doing these conversations these live streams reading papers having these discussions etc so my my model is updated through these conversations and then holistically we kind of come to a place where our models either converge through this communicative process on something that resembles each other's

models more similarly or not right we might have this divergence and then we go away with like this uncertainty like is my model wrong or or what was that argument about i mean everyone's been in conversations or discussions or heated scientific debates where you don't really come to a consensus right there is no general agreement or understanding and so then you walk away with maybe some some maybe my model is invalid and then do some more deep thought about it i mean that that's how i would that's how i would model it that's just me in the viat can at all 2020 paper with systems thinking we use this font framework formal documents ontology narratives and tools and so one kind of fun parallel or a way that conversation and the real dynamics that arise can be modeled in active inference is like agree to disagree two individuals can have high precision and convergence on a narrative inference we're agreeing to disagree they can both be precise and agreed on that and then at the first level they can literally disagree that's the whole point so that is modeled in active inference as nested models just like maxwell brought up and so we have nested models and the second order matrix can be the same between two agents yet their first level matrix can be very different and none of those matrices are the system map territory mel andrews acting 14. so we've come back to these ideas and we'll continue dance around them many many times because they're so fundamental that sometimes they're hard to see play out so steven and then anyone else yeah i think we have to be careful to note that it appears as if um information like what active inference tells us it appears if knowledge is coming in and that it's being we see it in our heads so to speak and yet the this conversation is is we're we're tuning through the action of our info niche our semantic niche of talking and we're aligning what our actions give us in our externality be it through the diagrams be it through this document and the slides or whatever the boundary objects are that we can align to given us partly due to our own folk psychology over the last 150 years the sense and this thing is really hard to counter this because this is kind of the way we tend to think you know that we have it in our minds i so to speak so it's true there's like this generative model yet at the same time the generative is also part of what we're doing we're generatively modeling each other's conversation and getting a sense of how i should say my actions of words none of those sentences were pre-made in my head you know they it's an action policy that's close enough to my sense-making so that it feels like i'm on the right track right and until someone looks at me really strange this is the um i'll carry on and i've been in these conversations a few times a bit more where how to keep doing right but i think that's something that's quite useful to when we talk about this modeling is oh we're modeling the action policies that are externalized and our generative model is as

good as it is at being able to enact um through participatory sense making what should happen next more than it is about having something that we've received in our generative model thanks steven that speaks to the interactionist versus instructionalism like is the fep model that blue is describing this generative model of the fep that's um in or embodied by all of us is it like a skyscraper made of glass and piping and there's 90 degree angles or is it some other type of metaphor that or even transcending metaphor and only realize through certain sequences of interaction so that's um really interesting and uh cool yeah there's great comments dave's uh throwing some biblical references in the chat and someone wrote i'm here first time in the live stream it's so good to listen all the cutting edge discussion about fep keep up the good work so nice work stephen and blue but thanks also for joining you have tuned your regime of attention hopefully you're having some questions and thoughts come to mind just as they are for us we're just on the other side of the video chat window and the participatory sense making side and how to mobilize that and enact it we can have participatory sense making with people on the video stream those who make asynchronous edits and improvements to these slides like we still have to go through blues prior thoughts that were added to um this slide deck and then from a history of science perspective and sociology like it's happening in a new way one thing that what was impressed on us i guess months to years ago forever ago in crypto active was like the time scale of these discussions going from peer review and annual conferences in person in annual or multi-annual cycles to like the pre-print and then we send the twitter message to the author of the preprint could you give a guest stream and then the authors can respond like looking at what maxwell wrote which was very important or said like bayesian mechanics that was live stream number 26. so we did a zero contextualizing video respecting that people have many different backgrounds approaching the work from and then two of the commentaries beel and um also aguilaria aguilara were in guest stream number five and seven so it's like we've heard from many of the important and that's this participatory scientific sense making were all part of a bigger positive project that's transdisciplinary it's not even just happening in the scientific literature but it has to a large extent been because these are research research advances these are conceptual theoretical advances and now we're all learning by doing and increasingly doing and applying and so there's so much to learn and do so um blue and then stephen and then there's more great questions in the live chat so go ahead you're like giving shouts out to us but i just want to like thank daniel for all the scaffolding that he's provided in the actin flab like to give us these um kind of the baseline for these discussions to take place um and also uh just to kind of um go back to what stephen was saying about you

know about modeling and enacting um really like i don't see knowledge as coming in right like when i'm in the in the modeling sense like i don't know that what you have is knowledge or what daniel has is knowledge or what the paper contains is knowledge um all of that in my world is and i i think that there's a difference between knowledge and information and so like when i'm interpreting comments by maxwell or steven or daniel when i'm interpreting that i take that information and transform it through my model right like so so some of what like take it or leave it or take what you want from this discussion and it's always like that i mean we always do that whether people have to remind us you know to do it or not right like you take away the parts that validate your model right or that improve upon your model or that you believe improve upon your model in some way so so then when i have it and i i have transformed it then maybe i can turn it into knowledge but i don't see coming in anything coming in as knowledge that's just me and with video chat it becomes super real like when there's lag all of a sudden it's like you're at the semantic level and then it's like wait just the syntax i'm not hearing the words and so it's like we dip into this lower level and then imagine if the audio is working fine you can totally hear non-equilibrium steady state but then conceptually what does that mean so these nested and overlapping layers of meaning so that's just very cool so steven and then there's a question in the chat yeah this is a good point blue and i suppose it's there's also this sense there's always got to be some belief in what you think the other person's saying like for instance i believe the words mean xyz because i know the english language right if we're all speaking in the language didn't know so i believe this word means this i hear this word these words together mean this so there's there's always and i'm hearing it seeing it i'm feeling it the fact that my chair is comfortable changes things right um there is this um sense that you know they talk about this in free energy principle is like it can also appear like you know so things as well as there being bayesian influence that you can appear like there's a bayesian influence type process which is what's happening my models can be updated and i think what i'm really interested in is for some of these things to then to be capitulated as knowledge i think there has to be and this is where the 4e work comes in is we bring this into some subconscious embodied meta well embodied practices which we can really tap into by sort of eliciting what they're like maybe getting a metaphor it feels like the knowledge is coming from around me and i'm grabbing it some other people it feels like it's coming in as a sequence of very clear things so my way of assembling connections of my belief structure to sensory information coming in will capitulate my modeling as best i can i'm going to be then trying to attune to the way everyone else is and we're all i'm sure it'd be

different if we're all standing up during all of these talks we might actually interpret and process a bit different if we're all in a pool having a cocktail you know i don't know i think blue's on a stationary bike about half the time i have my stationary bike hidden out of you but let's ask a question yeah exactly we could do a ride stream um we could do a pull stream but you gotta come to my house cool i remember that so let's ask a question from the live chat and then we'll return to some of blue's previously phrased questions about the paper and that's what's kind of fun like in the dot zero we get a lot of the details and the bulk of the paper and all the key ideas and figures if there are them and then in the dot one there's so many directions to go and i remember earlier it's like wait two times two hours group discussion on a paper i'm used to academia papers being discussed for one hour if ever and people having very heterogeneous understandings or they did or didn't read the paper so we find that there's always enough questions when we open the doors of the conversation to just like super fascinating direct directions for the discussion so again we'll ask a great question in the chat and then we'll kind of turn to some of these one questions because next week hopefully we'll have one or both of the authors to be talking with so joseph wrote how has active inference changed the way you view the world in daily life that's a big question do either of you want to go first or i'm happy to go i yeah i can chip in if you well one of the ways i've actually found it really helpful and i thought is um changing the way i think about how i see the world and also a bit change that i think about playing soccer so the first one is there was a there was this sense that i i i'm seeing the world right um and it's all a bit separate from me with active influence it it can be quite nice to go out and think about actually perturbing the world and feeling the world as if you're feeling the world with your eyes club touched touching the world with your hands seeing the world and touching the world with your eyes and i think that actually can help bring me out of my head so to speak in a metaphorical sense so i found that useful actually just thinking about that the world as being something that's out there and it's very much something i'm sensing and uh so that takes a little bit of the modernistic kind of concreteness and then with soccer as well it made me realize i used to always think you see those things on telly like you know like someone's kicking a ball you know like the equation's a bit like um terminates you know but now it's like well actually it's it's more like i'm just i'm just getting whatever my feeling is right and that whole other stuff it doesn't matter so then you realize all these really great sports players the really the ones that you they're they're actually they they've they've acclimatized to what it means to like methody must have this feeling for what it means to kick he doesn't need to have the trajectory of the ball in his head

so anyway i don't know if that helps us go through my crazy head blue you want to go or i have okay so there's a few angles and this could definitely be a worthy topic for another discussion but just to give a few angles so first thinking how we view the world that was a visual illusion in the original and so it reminds me of the development of thought from seeing the world that's the signal processing paradigm photons are pouring in and we're seeing categorizing classifying the world okay then we have predictive processing and that's like well you know our visual field for most individuals they see color in their entire visual field and there's equal clarity in their visual field even though there's an extremely heterogeneous distribution of resolution and color elements in the retina and we have a big old blind spot so color clarity of vision and blind spot suggests that we're experiencing a generative model a predictive model a vision that's predictive processing and it applies also to other sensory modalities and then that third step was where stephen took it with the active vision the epistemic foraging and we know that even though we don't directly experience these either there's eye saccades which move to the most informative part of the visual scene so to me it's about moving from signal processing 1.0 to predictive processing to active sensing and exploration as informed by epistemic and pragmatic value together you know eyes on the ball but you got to also see more than that generate more than that so that's one really important part the second one i'll also parallel exactly what steven said which is like it made me think even though i don't know about soccer i don't know about the tele wrong continent but it's kind of like the top down the video game model where there's like i see it more and more in sports replays where it's like it will pause and it will look like a video game but that's not the action model of the player on the using much more of a personal flow and a feel and interactions and cues and so they don't need to be having this 30 000 foot view they need the parry personal perspective so that is one piece and then just one a third one but i think another there's many more we could list would be like the reward oriented worldview so reward maximization is the paradigm reinforcement learning is how neurons associate utility maximization is what investing is about in contrast with an uncertainty reduction paradigm a precision optimization paradigm where qualified levels of precision of nested models allow a lot of flexibility and a lot more realistic phenomenology for different models because precision pursuit on an optimistic world model gets you utility so i just think we get utility from a precision first worldview but from a utility oriented worldview we don't always get precision and that i personally connect to how if we have like a tetrahedral based geometry of nature we get cubes but if we start with cubic geometry we never get tetrahedra so that's sort of where i connect it

utility is like bigger and a more general framework because utility falls out of precision on an optimistic generative model but merely having utility it's always going to be ad hoc or secondary where does curiosity and epistemic gain and all these other features of real decision making so those are a few of the key pieces how it changes how i've seen the world blue besides from like having integrated fully into like my daily vocabulary which is like you know we talk about things like the generative model like you know in completely unrelated offline conversations um so aside from like you know enhancing my vocabulary and building this new language of you know the fvp um that which is that's weird right like okay um but aside from that you know one of the things one of the recurrent um themes for me in active inference has been um the idea like we we touched on it in the mental action paper that we did like daniel you're gonna have to pull the pull it up for me because i can now remember you've got a better track you've got your model is updated and configured with all of the appropriate live stream numbers more more so than mine it's 28. it's 28 and it's only because of my extended niche but continue that's it and and so can you also query your extended niche because i can't ever remember the one where we talked about anxiety like was it if it was big five or sophisticated active inference or i always ask which one that one was but i can't ever that was a long time ago uh um but but going back you know touching on mental action and meditation and the idea that we touched on way in the long ago live stream of anxiety representing you know trying to project too many time steps ahead into the future like and having this relate to anxiety and and then like depression is relating to you know fixating too many steps in the past behind you right like so this like model of like neuroatypical things it just always reminds me to come back to the present moment and i'm looking for an excuse to do that anyway just because of my inclinations toward eastern religion and so forth so but but the reminder to stay in the present moment has now been like made through an appeal to my logical brain like outside of a spiritual sense and so that that kind of is is a cool you know effect of of active inference i think awesome there's so many fun questions in in the chat so i mean i'll i'll ask these and then we can we can we'll still have time to return to the paper so mihail asks okay one if you were to let your imagination roam freely where do you wish or see the best possible real life application so for that question that's a question for us and the community and beyond so we can maybe each just mention one thought that we're having but you know if you see a possible future state active inference of x let's act infer serve let's act first update our models and serve and make the impact that we want to see so this is actually an affordance of a question because if two people are like wow we should do this or could this

be useful if you can state it you're like 90 of the way there and we hope to develop the um transdisciplinary tools like active inference agent software and the expertise tools people and ideas and the education so that when somebody else who's not on this live stream there's just three of us has that insight or sees it being applied in a we want to help you scaffold that action trajectory but blue and then stephen so i guess i always come back to this too something that fascinates me fully is how information is integrated across scales so how is the model of of my you know dna strands imported into my cells imported into my tissues imported into my organs imported into my sense of self right like so how does this and how do i import into the organisms that i'm part of like the active inference lab and so forth so for me really like concretely realizing a mathematical way find information passing up and down like is it salience or what else what other way this emergence like for me that's like it like if active inference can nail that and this is like a fantasy daydream and i know people are working towards that but it's out there in the future but if i could really wish one thing if i could you know daydream about my ultimate fantasy of active inference being fulfilled it's completely that okay awesome modeling nested biological systems something that we've seen qualitatively discussed in the literature like ramstead at all 2018 answering schrodinger's question variational ecology variational neuroethology a lot of contributors to these works but that kind of a model would be epic stephen yeah i'm really interested i think active inference it doesn't have to do everything to be really a contribution it can it can it can fill certain gaps as well or give certain mechanisms that can give an insight either to tractably give away to model or a way to just think about things so what i'm curious about and i'm working on something called conceptual action sociometry which is a an action method um i'm trying to i'm really interested in how to work back from a high complexity context which would be too many dimensions to model from first principles so rather than modeling the generative model per se i'm looking at how to create an instrument track the info niche be inscribed by the agent so it's almost like a trace i'm i'm really interested in how we can go both ways as well as the sort of general sort of principled work that can be show how generative models are happening and being applied across things like context of autism and other areas where there's different types of classifications i'm really interested in kind of contextual work which will be outside of the ability to do full research but might be able to do an inquiry process and how to sort of just get a grip on that inquiry so um be it people looking at their lived experiences and things like that and we're actually talking about that in the tools section people are sort of giving me some advice or open to that once we get to the next phase awesome thanks for also sharing that and

then i think the system just the one that i'll mention would be online teams or research knowledge inquiry and doing scholarship and just at this point in time just facilitating the reduction of our uncertainty on the history of active inference and free energy principle in a few other areas because um tools for research and sense making would be really awesome for a lot of different reasons so cool and then the second question um from the hill in that sequence was what if we model our group conversation where we as humans disappear and where we are not exploring the universe but the universe is exploring itself utilizing us a beautiful thought question and there's a lot to think about that so blue yeah i mean i think that this uh possible right and we've talked about it in the computational boundary of the self the mike levin paper like ultimately it's like dissolution of the cognitive boundary and is there some like meta organism this goes to like the gaia hypothesis maybe not the universe but the earth or maybe the earth is part of the universe maybe there's some ultra ultimate like super organism that that cognitive boundary contains all of us and so just in the same way that like our cells are working towards our optimal function given the circumstances that they're they're presented with right like given that assumption maybe we're also working toward the optimal state of the universe like every interaction that occurs between all of us is working toward the optimal state of the active inference lab or the optimal state of the united states of america or the optimal state of you know scientific research or who knows right so any kind of umbrella category that we might fall into maybe we're serving the optimal state of that and really like i think quite possible and a really cool way to model active inference for sure another take on that from bucky fuller is this function first mentality led him to ask well what is the function of human and are we local sensemaker what is our role so those are big important questions let's return to 31.1 and kind of prime ourselves for our dot two because in the dot zero that's where like whoever's really excited about the paper maybe whoever contacted the authors or maybe one of the authors themselves or just a few people who want to learn by doing the dot zero which was like blue and i for this one 31.1 the previous video we go through the paper the dot one we unpack that's sort of the bathtub phase everything is getting spread out a lot of sociological criticisms or uh insights a few random questions from the get from the guests of our stream a lot of different areas a lot of threads get pulled out and then in the dot 2 we try to have some synthesis because it's not the end of the discussion with the dot 2 but we have to move on from a live stream perspective so the dot 2 especially when the authors can be there we can get some more insights we can resolve some questions that arose during the dot one and dot zero and the inter uh meaning times so questions that didn't arise before we enacted

this process and then we get to ask those and then kind of use that as our jumping off point as we move on to our next phase so here in the dot one at the end it's kind of like a fractal.² we're kind of trying to tie a few things together returning to the text as a strange attractor so that we'll have a whole week to read think sleep stationary bicycle mind and body relax maybe do our day job if we can find any and um be here you know next week or whenever works to continue the discussion so blue um in this questions for 31.1 what were some quotes or some questions from the paper that you wanted to like learn more about or ask the authors about well just as i mentioned um you know even earlier in this live stream today the dynamic equilibrium of living systems like i'm not understanding that at all like maybe that's homeostasis but homeostasis is not in my mind dynamic equilibrium and i think that this um you know bayesian mechanics for stationary processes and the idea of a non-equilibrium steady state density uh kind of supersedes this concept of dynamic equilibrium because homeostasis is not equilibrium and so that's kind of a fundamental like it like bends in my brain a little bit when i when i've been reading over this paper um and also uh you know it says here we should pay closer attention to fundamental differences between physics and biology um and how these indifferences interact with importing concepts from physics into the modeling of biological targets so i mean is it physics all the way down because i always thought it was physics all the way down right like so i don't know that goes to um you know like when i was in school they always said like you know biology is not as precise as chemistry and chemistry is not as precise as physics and physics is not as precise as math right like so there's this like tiered hierarchy of how you know fundamental your study can be right and so it i think biology is based in the concept of physics but but also chemistry i feel like was really omitted and left out and like in my heart i'm a chemist uh stephen i'm sure can relate to that um but really like so so that's kind of one other uh thing i wanted to explore and then um the other part of um of the questions that i have for the authors are you know in active inference are the active states internal um because they said in internal states include active states but i have always thought of active states and sensory states as part of the interface between internal and external states as part of the markov blanket and so i just wanted to clear that up maybe if that's a misunderstanding on my part or i mean i guess action can happen internally like as we've seen in the mental action so that's definitely possible but but not all active states are part of internal states i thought so i don't know maybe that's some something that we can tease apart with the authors and i'm looking forward to having them here and asking them these three questions which are the ones that kind of for me um you know don't

align with my current model of the fbp they were salient they were different from your generative mall so it's a nice way to put it and then i'll just show this one comic and then steven so this is a classical uh xkcd comic the field's arranged by purity okay so it's a comic you know retweets are not endorsements but you have uh this is a classic disciplinary hierarchy slash ponzi scheme pyramid scheme whatever you you have uh the physicist let's just start in the middle it's nice to be on top that's the the common pinnacle of the pyramid in certain paradigms and then like the physicists can make claims about lower levels but then if the psychologist makes a claim about physics hey we need a sociology of molecules it'd be like that's not science but then the physicist talking about sociophysics is like wow that's so modern and exciting so there's a sort of um linearity as well as potentially some sociological scaffolding that facilitates the diffusion of information and expertise and uh notoriety like one way but it's not always a two-way street and then the joke i guess in the xkcd is like the mathematician saying hey i didn't see you all from over here so again just a comic now playing with the media what does it look like to i don't know close the loop or have these learn about each other's perspective or there's a million ways that we could go counterfactual on this xkcd what if i mean that's the name of the xkcd's author's book which is kind of fun um but how does active inference challenge this either specifically or just how are broader ongoing sequences of actions decentralized science desai open science open access open source global collaboration transdisciplinary application complexity science i mean there's a million ways in which this is kind of funny because it speaks to an academic hierarchy and maybe we've even met some characters who really embodied these archetypes but what are the archetypes going to be going forward what are more accessible archetypes and ones that are more true to form and they can still be funny i'm sure we'll have acting memes just enough to warm the hearth but this is just one view that kind of speaks that disciplinary hierarchy so yeah steven then blue yeah i think this this is where this foundational shift is this this desire for purity um starts has been really challenging of course now we're in this post-normal science era where you know the reality is that we're gonna have to make decisions and not just wait for the purity of the the random control trial or whatever it is to make choices so how can we be more systematic and rigorous and so then complexity comes in and the challenges that we are facing is that a lot of psychologists and psychiatrists um have become i've also gone down the purity route in a way they're trying to like the idea is they're all trying to be more pure and the mathematician can be more pure than the others but that question about being more pure i think this is what blues points make a really good they come into question like physics comes

into question chemistry comes into when you talk about far from equilibrium states even the idea of it and in chemistry they call it far from equilibrium chemistry in physics they call it non-equilibrium states right um because they can actually method and the mathematicians would call it lorenz attracts estates right and but in either case you know you only have to look at lorenz attractor the whole point is you don't know where you are in it right and and not in the same way as quantum well that's the question whether you when you get into the fine details we don't know yet but it's still different to quantum right and i think that's where that quantum contextuality comes in as well so anyway what's your thoughts on that the thing that just came to mind and then blue is like this is a linear right it's open-ended um but sociologists and philosophers make important insights about the philosophy and history of science and so the mathematician can say yeah we're the most pure or we're the most correct or whatever it happens to be but they're nested in a sociological context in a historical in a philosophical context so it is closed from a causal closure perspective because it's not like physics have just broken out of the frameworks of thought how to move this line a conversation that's including all of these disciplines in a trans discipline and including people who aren't in academia or research in that as well blue this is an interesting comic and i i usually like don't even talk about psychologists or sociologists because that's not science right no i'm just kidding um i really actually have a lot of respect for psychology and sociology um but it's just not something that's ever been like lumped into my pyramid before so uh thank you for for sharing that and seeing it in this way like maybe they should arrange by like the fundamentality of it like sociology without like social interaction we would not develop things like mathematics right like there would be no need so so in terms of what's more fundamental like maybe sociology is fundamental and psychology like we're not even thinking about what makes us work if we're not psychologically okay right like if we're in a bad psychological state nobody's thinking about intra cells and dna like that's there there's just no you have to have like the fundamental needs have to be met before you start to ponder things like what am i made of and how do my parts work right and and really like so so maybe in terms of fundamentality and then we are biology and and then you know so maybe in terms of the fundamental things sociology is the pinnacle of that again there's probably many ways in which different perspectives or active inference can challenge this or at least kind of continue this joke a little bit but it's pure active inference it's really about sociological context we really have a mathematical formalism it's really grounded in the ways that people have been studying physics that's one of the questions of 31 the paper is to what extent is it has it could it been

grounded in physics but we really have physics and we really are talking about biology and for comprehensiveness and and getting to the fundamentals it is really fundamentally relational so we're not taking this reductionist purity spiral we're taking like a transdisciplinarity purity and we're just saying this is our keyword and to really get out what we're talking about we're really going to respect all of these perspectives so that's kind of a funny challenge that you wouldn't necessarily hear arising from a disciplinary conversation so stephen go for it yeah i think this is also one thing you know we say though that we see the importance of social interaction social inference and um the challenges that i think actually this might be quite apt for where we had at the beginning with maxwell's conversation is the perspective of the sociologist becomes what is ascribed to social interactions you know the psychologists is what's ascribing the mental models that we have in our head the biologists are ascribing the categories the chemist is describing the steady state you know when you get into this agent-based modeling which again i think is more familiar to the newer work in biology but it's something that i had to start to learn about you know and looking at these dynamics and modeling actually starts to it starts to go under the ground of all of these in it doesn't make them invalid because they they are useful when you can take a perspective on the systemic work but we're not taking perspectives in this and there's that's the whole ground starts to become shifting um and that's really interesting okay here's another kind of a subtlety actually so um sociologists okay just starting there that's our prior you know d coming in from the left sociology is applied psychology okay so we're going all the way you know chemistry is applied physics it's just like it's interesting the conversation is going that way i'm just seeing another comic where it could have been like physics is applied math chemistry is applied physics like who is saying which field is an application of which other fields so there's a lot of ways that conversation could play out and by imagining the counterfactuals the semantic the rhetorical counterfactuals and then asking well what is concordant or discordant with our generative model and why that's a very rich starting point for transdisciplinary discussions because the whole inter meta transdisciplinary discussion it's been decades centuries truly with different words so how are we going to look at this with a new light and maybe even integrate some of the truly recent developments in tools like performance computing and interactive notebooks and ontologies these didn't exist one or two hundred years ago in the same way they do today respect qualitative and quantitative and different backgrounds recognizing that we're never going to get the end of the story even if you got 100 phds in any one of these fields or lived 100 years in a field outside of academia you still wouldn't be at the end so how are we going to take a

radically different approach learning and applying maybe active inference could have something to do with it in the last few minutes anyone else in the live chat could just ask a final question otherwise let's just um sort of recap the big aims and claims of this paper were about the foundations in statistical physics and dynamical systems theory for the free energy principle and what's on the line what's on the table for whether the theoretical underpinnings are adequate or not for fep and active well it's the practical utility as well as the theoretical adequacy that's what matters they claim that their work is starting to fill in this gap and whether we think that there's any any other pebbles in that gap or the gap is uh you know a meter across or a kilometer across there is a differential and we can all positively contribute to that through discourse and through contributions to the literature and then the main argument of the paper and one of the big ideas that'll be really cool to talk about is that the foundations in physics allow the fep models to be maximally general to describe across systems but in the author's words these assumptions also decrease the ability of fep models to include enough factors to provide biologically plausible representations for living systems and the way that they are in the world so you know big idea physical and physics grounding of fep and what that says about the theoretical adequacy as well as the practical utility and then this idea of generalization and if we over generalize do we sort of leave behind some of the features that we wanted to explain in the first place so that's a little bit of the recap and i'll just pull to our kind of closing slide and ask these questions and then stephen and blue you can give a last thought what would a good understanding enable what are the unique predictions and the implications of this research or of what you thought of after learning about this what are the next steps for fep and what are the goals of this or your research and then what are you still curious about what is your intrinsic motivation that's going to keep you reading these papers and contributing to the discussion so stephen or blue any last thoughts stephen first and then i think in terms of this paper anyway i think that this really does highlight this need to understand the difference between particles voxels things systems i think this because these types of critiques keep coming up right at times i know it's it has a risk of kind of tying the free energy principle community in some knots um and uh so i think that it's a challenge and i think that that that may be a part of where this goes and at the same time i think it's a good provocation and there is some questions here about how and why those confusions arise so i think that might be uh um you know i'm curious about learning more about that because i think that's going to help in other ways in doing this work thank you stephen blue it's awesome to me i think that i'm talking about the um general general generalizability sorry that didn't

come out well the generalizability and applicability of like the the free energy principle i i really like that that's also like a trade-off between like epistemic and pragmatic value like it reminds me of that balance that we see in the fep and and so i think like the ultimate goal is to resolve that right like how do we strike the balance between epistemic and pragmatic value how do we strike that balance between generalization and application great point we want to describe many systems and describe them well and walk in with a scaffold but minimal constraining priors that prevent us from seeing some fundamental or even essential attributes of systems and then that general in specific connecting that to learning and doing to epistemic and pragmatic values or gains which hint we have included in the actin formalism as part of our policy selection approach so stephen and blue plus everybody who asked truly excellent questions in the live stream will see you in an organizational unit meeting or the discord or some other way just this is your affordance to participate and contribute so thanks again to everybody and we'll see you next time bye